MIL-E-1/1638A
16 June 1970
SUPERSEDING
MIL-E-1/1638
20 March 1967

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 2K22

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: *Reflex, tunable frequency range 4, 240 to 4, 910 MHz, rated power output 100 mW, integral cavity*

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	Ehk	T (anode)	Eb
Unit:	V	Vdc	Vdc	mAdc	Vdc	°C	Vdc
Maximum:	6.8	350	-350	37	50	110	330
Minimum:	5.8	---	0	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Base:	B6-90
Cathode:	Unipotential, oxide coated	Cap:	C1-4
Mounting position:	Any	Weight:	1.8 ounces (max)

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	VSWR	F	
Unit:	V	Vdc	Vdc	---	MHz	
Test 1: Mode 3-3/4						
Maximum:	---	---	-180	1.10:1	4,589	
:	6.0	300	---	---	4,575	(see note 2)
Minimum:	---	---	-120	---	4,561	
Test 2: Mode 3-3/4						
Maximum:	---	---	-180	1.10:1	4,589	
:	6.3	300	---	---	4,575	(see note 2)
Minimum:	---	---	-120	---	4,561	

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	---					
4250	Power output	---	1	Er/max Po	Po	75	---	mW
4214	Cathode emission	---	---	Ef = 6.3 to 5.8 V; t = 2 minutes	ΔIk/Ik	---	15	%
4223	Mechanical tuning range	---	2	Er = -155 to -235 Vdc Er = -75 to -155 Vdc	F F	4,910 ---	--- 4,240	MHz MHz
1256	Electrode current (cathode)	---	2		Ik	---	32	mAdc
4229	Total reflector current	---	2		Ir	---	-7.0	μAdc
4229	Reflector-leakage current	---	2		Ir	---	-5.0	μAdc
4229	Reflector-gas current	---	2		Ir	---	-2.0	μAdc
	Quality conformance inspection, part 2							
1301	Heater current	---	---	Ef = 6.3 V	If	410	470	mA
1336	Heater-cathode leakage	---	---	Ehk = ±45 Vdc	Ihk	0	100	μAdc
1031	High-frequency vibration	---	---	10G; t = 120	---	---	---	---
1211	Insulation of electrodes	---	---	300 Vdc; tube cold	Rk - rs Rf - rs	2.0 2.0	--- ---	Meg Meg
4280	Electronic tuning range	---	---	F = 4,575 ± 14 MHz; Er/50% max Po	ΔF	25	---	MHz
4231	Electronic tuning hysteresis (1)	---	2	Er/max Po	Ratio	---	5	%
4223	Tuning rate	---	---		Turns	---	3.5	---
	Quality conformance inspection, part 3							
---	Life-test provisions	---	2	Group C or S; Er = -350 Vdc	t	500	---	hrs
---	Life-test end point	---	---					
4250	Power output	---	1	Er/max Po	Po	25	---	mW
	Periodic-check test							
4027	Temperature coefficient	---	---		ΔF/°T	+0.05	-0.10	MHz/°C

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

2. The tube shall be fixed firmly in a suitable socket by clamps which make contact at the base and adjacent platform only. The measurements on the tubes in an oscillating state shall be made with the output line coupled through a connector into a 50-ohm lossy line operating into a load adjusted to 50 ohms at 4,575 MHz.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

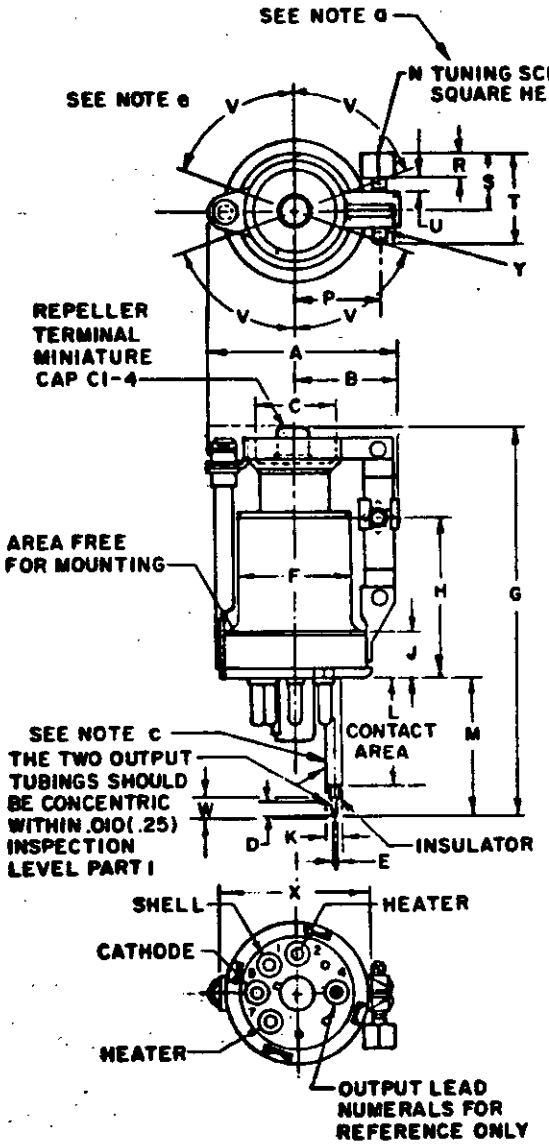
User activities:
Army - ME, AV, SM, WC
Navy - AS, OS, MC, CG, SH
Air Force -

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2411)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note d)		
H	1.313 (33.35)	1.437 (36.50)
J	.330 (8.38)	.412 (10.46)
K	.135 (3.43)	.145 (3.68)
L	.876 (22.25)	.985 (25.02)
M	1.188 (30.18)	1.250 (31.75)
P	.657 (16.69)	.719 (18.26)
W	.172 (4.37)	.204 (11.10)
Quality conformance inspection, part 2		
A	1.531 (38.89)	1.609 (40.87)
B	.812 (20.62)	.859 (21.82)
D	.156 (3.96) Nom	
F	1.000 (25.40)	1.016 (25.81)
G	3.250 (82.55)	3.500 (88.90)
N	.182 (4.62)	.192 (4.88)
R	.203 (5.16)	.235 (5.97)
S	.484 (12.29)	.516 (13.11)
U		.130 (3.30)
Y	5-40 thd	
Periodic-check		
C	.767 (19.48)	.797 (20.24)
E	.035 (.89)	.045 (1.14)
T	.665 (16.89)	.797 (20.24)
V	70°	
X	1.271 (32.28)	1.312 (33.32)

FIGURE 1. Outline drawing of electron tube type 2K22.

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NOTES:

a. The tuning screw shall be lubricated with oildag or equivalent noncorrosive lubricant. It shall be capable of being operated smoothly through its entire range without perceptible binding.

b. This base shall be capable of being inserted in a gage 1.218 (30.94 mm) thick having 4 holes .250 (6.35 mm) deep from the top of gage whose diameters are .103 (2.62 mm) for the contact pins. Remaining portion of hole to be clearance. Approximately .015 (.38 mm) larger in diameter and a fifth hole whose diameter .160 (4.06 mm) x 1.218 (30.94 mm) deep for the output lead. All holes located on the true centers. Locate also a center hole having the contour of the pilot but with a clearance of .002 (.05 mm) over the maximum diameter.

c. Nickel tubing (optional silver plate) or steel tubing 150 MSI min. silver plate. (Quality conformance inspection, part 3 (periodic-check).

d. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

e. V area free for mounting.

FIGURE 1. Outline drawing of electron tube type 2K22 - Continued.

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